

Chapter 16 Event-Driven Programming

Section 16.2 Event and Event Source

1. Pressing a button generates a(n) _____ event.
 - a. ItemEvent
 - b. MouseEvent
 - c. MouseMotionEvent
 - d. ActionEvent
 - e. ContainerEvent
2. Clicking the closing button on the upper-right corner of a frame generates a(n) _____ event.
 - a. ItemEvent
 - b. WindowEvent
 - c. MouseMotionEvent
 - d. ComponentEvent
 - e. ContainerEvent
3. Which of the following statements are true?
 - a. If a component can generate an event, any subclass of the component can generate the same type of event.
 - b. All the event classes are subclasses of EventObject.
 - c. A component on which an event is generated is called the source object.
 - d. Every GUI component can generate MouseEvent, KeyEvent, FocusEvent, and ComponentEvent.

Section 16.3 Listeners, Registrations, and Handling Events

4. The component that processes the listener is called _____.
 - a. the source object
 - b. the listener object
 - c. the adapter object
 - d. the adaptee object
5. Which of the following statements are true?
 - a. Each event class has a corresponding listener interface.
 - b. The listener object's class must implement the corresponding event-listener interface.
 - c. A source may have multiple listeners.
 - d. The listener object must be registered by the source object.
 - e. A listener may listen for multiple sources.
6. Which of the following statements registers a panel object p as a listener for a button variable jbt?
 - a. addActionListener(p);
 - b. jbt.addActionListener(p);
 - c. jbt.addActionEventListener(p);
 - d. jbt.addEventListener(p);
7. The interface _____ should be implemented to listen for a button action event.
 - a. MouseListener
 - b. ActionListener
 - c. FocusListener
 - d. WindowListener
 - e. ContainerListener
8. The method in the ActionEvent _____ returns the action command of the button.
 - a. getActionCommand()
 - b. getModifiers()
 - c. paramString()

- d. `getID()`
- 9. The handler (e.g., `actionPerformed`) is a method in _____.
 - a. a source object
 - b. a listener object
 - c. both source and listener object
 - d. the `Object` class
 - e. the `EventObject` class
- 10. Every event object has the _____ method.
 - a. `getSource()`
 - b. `getActionCommand()`
 - c. `getTimeStamp()`
 - d. `getWhen()`
 - e. `getKeyChar()`

Section 16.4 Inner Classes

- 11. Which of the following statements are true?
 - a. Inner classes can make programs simple and concise.
 - b. An inner class can be declared public or private subject to the same visibility rules applied to a member of the class.
 - c. An inner class can be declared static. A static inner class can be accessed using the outer class name. A static inner class cannot access nonstatic members of the outer class.
 - d. An inner class supports the work of its containing outer class and is compiled into a class named `OuterClassName$InnerClassName.class`.
- 12. Suppose A is an inner class in Test. A is compiled into a file named _____.
 - a. `A.class`
 - b. `Test$A.class`
 - c. `A$Test.class`
 - d. `Test&A.class`
- 13. Which statement is true about a non-static inner class?
 - a. It must implement an interface.
 - b. It is accessible from any other class.
 - c. It can only be instantiated in the enclosing class.
 - d. It must be final if it is declared in a method scope.
 - e. It can access private instance variables in the enclosing object.

Section 16.5 Anonymous Class Listeners

- 14. Which of the following statements are true?
 - a. An anonymous inner class is an inner class without a name.
 - b. An anonymous inner class must always extend a superclass or implement an interface, but it cannot have an explicit `extends` or `implements` clause.
 - c. An anonymous inner class must implement all the abstract methods in the superclass or in the interface.
 - d. An anonymous inner class always uses the no-arg constructor from its superclass to create an instance. If an anonymous inner class implements an interface, the constructor is `Object()`.
 - e. An anonymous inner class is compiled into a class named `OuterClassName$n.class`.
- 15. Suppose A is an anonymous inner class in Test. A is compiled into a file named _____.
 - a. `A.class`
 - b. `Test$A.class`
 - c. `A$Test.class`
 - d. `Test$1.class`
 - e. `Test&1.class`

17. Analyze the following code.

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

public class Test extends JFrame {
    public void Test() {
        JButton jbtOK = new JButton("OK");
        add(jbtOK);
        jbtOK.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                System.out.println("The OK button is clicked");
            }
        });
    }

    public static void main(String[] args) {
        JFrame frame = new Test();
        frame.setSize(300, 300);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

- a. The program has a compile error because no listeners are registered with jbtOK.
- b. The program has a runtime error because no listeners are registered with jbtOK.
- c. The message "The OK button is clicked" is displayed when you click the OK button.
- d. The actionPerformed method is not executed when you click the OK button, because no instance of Test is registered with jbtOK.
- e. When you run the program, the button is not displayed, because the constructor is declared wrong. It should be declared public Test(), not public void Test().

Section 16.6 Alternative Ways of Defining Listener Classes

18. Analyze the following code.

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

public class Test extends JFrame implements ActionListener {
    public Test() {
        JButton jbtOK = new JButton("OK");
        getContentPane().add(jbtOK);
    }

    public void actionPerformed(ActionEvent e) {
        System.out.println("The OK button is clicked");
    }

    public static void main(String[] args) {
        JFrame frame = new Test();
        frame.setSize(300, 300);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

}

- a. The program has a compile error because no listeners are registered with jbtOK.
- b. The program has a runtime error because no listeners are registered with jbtOK.
- c. The message "The OK button is clicked" is displayed when you click the OK button.
- d. The actionPerformed method is not executed when you click the OK button, because no instance of Test is registered with jbtOK.
- e. None of the above.

19. Analyze the following code.

```
1 import java.awt.*;
2 import java.awt.event.*;
3 import javax.swing.*;
4
5 public class Test extends JFrame {
6     public Test() {
7         JButton jbtOK = new JButton("OK");
8         JButton jbtCancel = new JButton("Cancel");
9         getContentPane().add(jbtOK);
10        getContentPane().add(jbtCancel);
11        jbtOK.addActionListener(this);
12        jbtCancel.addActionListener(this);
13    }
14
15    public void actionPerformed(ActionEvent e) {
16        System.out.println("A button is clicked");
17    }
18
19    public static void main(String[] args) {
20        JFrame frame = new Test();
21        frame.setSize(300, 300);
22        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
23        frame.setVisible(true);
24    }
25}
```

- a. The program has compile errors on Lines 11 and 12 because Test does not implement ActionListener.
- b. The program has compile errors on Line 15 because the signature of actionPerformed is wrong.
- c. The program has compile errors on Line 20 because new Test() is assigned to frame (a variable of JFrame).
- d. The program has runtime errors on Lines 9 and 10 because jbtOK and jbtCancel are added to the same location in the container.
- e. None of the above.

20. Analyze the following code.

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Test extends A {
    public static void main(String[] args) {
        A frame = new Test();
        frame.setSize(200, 100);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```

```

    frame.setVisible(true);
}

JButton jbtOK = new JButton("OK");

public Test() {
    getContentPane().add(jbtOK);
    jbtOK.addActionListener(this);
}

public void actionPerformed(ActionEvent e) {
    super.actionPerformed(e);

    if (e.getSource() == jbtOK)
        System.out.println("OK button is clicked");
}
}

class A extends JFrame implements ActionListener {
    JButton jbtCancel = new JButton("Cancel");

    public A() {
        getContentPane().setLayout(new FlowLayout());
        getContentPane().add(jbtCancel);
        jbtCancel.addActionListener(this);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == jbtCancel)
            System.out.println("Cancel button is clicked");
    }
}

```

- a. The program displays Cancel button on the left of the OK button.
- b. When you click the OK button the message "OK button is clicked" is displayed.
- c. When you click the Cancel button the message "Cancel button is clicked" is displayed.
- d. If the super.actionPerformed(e) statement in the actionPerformed method in the Test class is omitted, no message is displayed if you click the Cancel button.
- e. All of the above.

Section 16.8 Mouse Events

21. To detect whether the right button of the mouse is pressed, you use the method _____ in the MouseEvent object evt.

- a. evt.isAltDown()
- b. evt.isControlDown()
- c. evt.isMetaDown()
- d. evt.isShiftDown()

22. To listen to mouse clicked events, the listener must implement the _____ interface or extend the _____ adapter.

- a. MouseListener/MouseAdapter
- b. MouseMotionListener/MouseMotionAdapter
- c. WindowListener/WindowAdapter
- d. ComponentListener/ComponentAdapter

23. To listen to mouse moved events, the listener must implement the _____ interface or extend the _____ class.

- a. MouseListener/MouseAdapter
- b. MouseMotionListener/MouseMotionAdapter
- c. WindowListener/WindowAdapter

d. `ComponentListener/ComponentAdapter`

24. To get the x coordinate of the mouse pointer for the `MouseEvent` `evt`, you use _____.

- a. `evt.getX()`
- b. `evt.getPoint().x`
- c. Either A or B
- d. Neither A nor B

25. The listener's _____ method is invoked after a mouse button is pressed (but not released yet).

- a. `public void mousePressed(MouseEvent e)`
- b. `public void mouseReleased(MouseEvent e)`
- c. `public void mouseEntered(MouseEvent e)`
- d. `public void mouseExited(MouseEvent e)`
- e. `public void mouseClicked(MouseEvent e)`

26. The listener's _____ method is invoked after a mouse button is released.

- a. `public void mousePressed(MouseEvent e)`
- b. `public void mouseReleased(MouseEvent e)`
- c. `public void mouseEntered(MouseEvent e)`
- d. `public void mouseExited(MouseEvent e)`
- e. `public void mouseClicked(MouseEvent e)`

Section 16.9 Listener Interface Adapters

16. Which of the following are correct names for listener adapters?

- a. `ActionAdapter`
- b. `MouseAdapter`
- c. `KeyAdapter`
- d. `WindowAdapter`

Section 16.10 Keyboard Events

27. To listen to keyboard actions, the listener must implement the _____ interface or extend the _____ class.

- a. `MouseListener/MouseAdapter`
- b. `KeyListener/KeyAdapter`
- c. `WindowListener/WindowAdapter`
- d. `ComponentListener/ComponentAdapter`

28. Which of the following statements are true?

- a. The `keyPressed` handler is invoked when a key is pressed.
- b. The `keyReleased` handler is invoked when a key is released.
- c. The `keyTyped` handler is invoked when a key is entered.
- d. The `keyTyped` handler is invoked when a Unicode character is entered.

29. To check whether a `DELETE` key is pressed or released, which handler should be used?

- a. `keyPressed(KeyEvent e)`
- b. `keyReleased(KeyEvent e)`
- c. `keyTyped(KeyEvent e)`

30. The `getKeyCode()` method of the `KeyEvent` returns _____.

- a. a character
- b. the ASCII code of the character
- c. the Unicode code of the character
- d. None of the above.

31. What is the value of `evt.getKeyCode()` or `evt.getChar()` for the `keyTyped()` events?

- a. A character

- b. The ASCII code of the character
 - c. The Unicode code of the character
 - d. VK_UNDEFINED
32. To enable a component to listen to keyboard events, you need to ____
- a. Implement the KeyListener interface for the component.
 - b. Override the isFocusTraversable method defined in the Component class to return true.
 - c. Invoke the component's requestFocus method to set focus on this component.
33. To be a listener for ActionEvent, an object must be an instance of ____.
- a. ActionEvent
 - b. ActionListener
 - c. EventObject
 - d. WindowListener
 - e. WindowEvent
34. You can use getWhen() to find the time of an ____.
- a. ActionEvent
 - b. MouseEvent
 - c. KeyEvent
 - d. ComponentEvent
 - e. WindowEvent

Section 16.11 Animation Using the Timer Class

35. Which of the following statements are true?
- a. You can add a listener in the Timer constructor.
 - b. You can use the addActionListener method in the Timer class to add a listener.
 - c. You can specify a delay in the Timer constructor.
 - d. You can specify a delay using the setDelay method.
36. Which of the following statements are true?
- a. You must always specify a listener when creating a Timer object.
 - b. You can add multiple listeners for a Timer object.
 - c. To stop a timer, invoke timer.stop().
 - d. To start a timer, invoke timer.start().
 - e. When a timer is created, it is automatically started.